2. 8-Queen Problem

Aim:

To write a Python program to solve the 8-Queen problem using the backtracking algorithm.

Algorithm:

1. Start the program.
2. Create an 8 × 8 chessboard.
3. Place a queen in the first row.
4. Check whether the queen is safe.
5. If safe, place the queen in the next row.
6. If no safe position exists, backtrack.
7. Repeat until all 8 queens are placed.
8. Display the chessboard.
9. Stop.

Code:

```
def solve(row, cols, diag1, diag2, board):
    if row == 8:
        return True

    for col in range(8):
        if (col not in cols and
            row-col not in diag1 and
            row+col not in diag2):

            board[row] = col

            if solve(row+1,
                     cols | {col},
                     diag1 | {row-col},
                     diag2 | {row+col},
                     board):
                return True

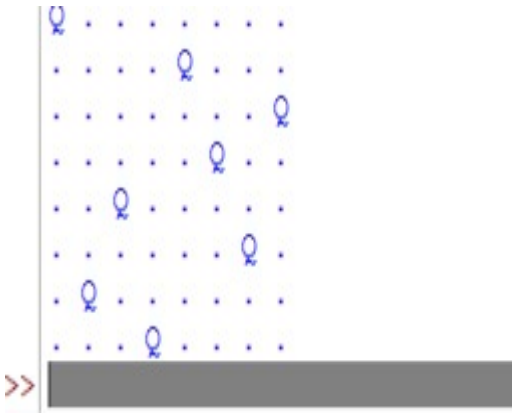
    return False

board = [-1] * 8

solve(0, set(), set(), set(), board)

for r in range(8):
    print(" ".join(
        "Q" if board[r] == c else "."
        for c in range(8)
    ))
```

Output:



Result:

Thus, the Python program to solve the 8-Queen problem was executed successfully.